Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is the difference between a control dependency and a data dependency? 1 / 1 point

- Control dependencies describe how the program flow is structured, while data dependencies describe how pieces of data affect each other.
- Control dependencies describe how the program flow is structured, while data dependencies describe how cache hit rates affect the program's operation
- Control dependencies describe how the program operation depends on the CPU it's running on, while data dependencies describe how pieces of data affect each other
- None of the above

Nice work

Correct. A control dependency governs how different sequences of instructions affect each other, while a data dependency governs how different pieces of data affect each other.

2. Which of the following is an example of a scalar optimization technique? 1 / 1 point

- Loop unrolling - performing every nth iteration of a loop, including the contents of the skipped iterations within the performed iteration
- Strength conglomeration - grouping together long, computationally intensive sections of code to make the rest of the program prettier to look at
- Iteration peeling - removing certain conditionals that would prevent parallelization from within loops
- Strength reduction - replacing a computationally expensive option with a faster one

Nice work

Correct. Not all operations have the same cost. When the goal is the optimization of code, choose expressions to reduce computational cost.

3. Which of the following is not an example of a loop optimization technique? 1 / 1 point

- None.
- Loop fission - splitting a complex loop into several smaller ones, to facilitate efficient cache usage and parallelization
- Unswitching - moving certain conditionals to avoid inefficiency in program flow
- Loop fusion - combining loops that run over the same range of indices

Nice work

Correct. Answers a-c are all examples of loop optimization.

4. Which GCC flag will likely be implemented if called to improve optimization in the following code snippet? 1 / 1 point

```
X=100
Y=200
DO i=1, n
   Z=X+Y
  var=g(i)+w(i)+Z
END DO
```

- funswitch-loops
- fpeel-loops
- fmove-loop-invariants
- ftree-loop-distribution

Nice work

This is the GCC compiler flag that deals with loop invariant code

5. What is one reason why optimizing loops with compiler flags results in faster run times? 1 / 1 point

- Large datasets are not able to be held in cache, so optimizing the code results in the code being split into multiple locations, which makes it easier to reference
- The code references main memory rather than cache, which is quicker to access
- Complex loop dependencies that are optimized by calling compiler flags can reduce these dependencies, resulting in faster run times
- The code references cache rather than main memory, which is quicker to access

Nice work

Code that resides in cache is the quickest to access

6. Why would it be more advantageous to perform Fourier transforms in Numpy rather than a compiled language? 1 / 1 point

- Compiled code is generally slower and less efficient than scripted languages such as Python
- Python code is a more modern language that has been designed with greater efficiency
- It's not necessarily faster to perform Fourier transforms in Numpy than in a compiled language
- Numpy arrays are faster because they can only contain one element type, resulting in a smaller memory footprint

Nice work

This is true – it will depend on the data, the knowledge of the programmer in each language, and other issues that aren't related to whether one language is better than the other

7. Which of the following is an example of a data dependency? 1 / 1 point

- A loop that iterates over values in an array
- A conditional statement that checks the value of a variable
- An assignment statement that uses the value of a variable from a previous statement
- A function call within a loop

Nice work

Correct. This creates a data dependency where the second statement depends on the value from the first statement

8. What is the purpose of loop unrolling? 1 / 1 point

- To reduce the number of iterations in a loop
- To increase the number of operations performed within each iteration
- To eliminate loop-carried dependencies
- To improve cache performance by accessing adjacent memory locations

Nice work

Correct. Loop unrolling combines multiple iterations into a single iteration, increasing ops per iteration.

9. Which optimization technique involves moving code that does not change within a loop outside of the loop? 1 / 1 point

- Loop fusion
- Loop fission
- Loop hoisting
- Loop interchange

Nice work

Correct. Hoisting moves loop-invariant code outside the loop.

10. What is the potential benefit of using NumPy arrays compared to Python lists for numerical computations? 1 / 1 point

- NumPy arrays can hold mixed data types, while Python lists cannot
- NumPy arrays are dynamically resizable, while Python lists have a fixed size
- NumPy arrays use compiled C code and can take advantage of CPU vectorization
- NumPy arrays have a larger memory footprint, which can improve performance

Nice work

Correct. This is a key performance benefit of NumPy.